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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

The purpose of this project is to analyze customer purchasing behavior using a Diwali sales dataset. Diwali is one of the largest commercial shopping periods in India, making its sales data valuable for understanding customer preferences, regional demand, and purchasing trends. (Kotler et al., 2017). The dataset contains demographic and transactional attributes supporting analysis of customer segmentation, spending behavior, and regional sales performance, consisting of 11,251 rows and 15 columns. The core objectives are to identify top performing states, evaluate the impact of demographics on spending, and uncover categorical associations between products and geographical zones. This project follows a structured methodology encompassing data understanding, preparation, cleaning, visualization, statistical analysis, and hypothesis testing. (Provost and Fawcett, 2013).

2. Data Understanding

Data understanding is the process of examining a dataset to identify its structure, variables, quality issues, and analytical potential before performing transformation or statistical analysis. This stage is crucial as it ensures that the dataset is suitable for generating accurate insights while helping identify issues such as missing values, redundant columns, and inconsistent variable types.

2.1. Data Source Information

The Diwali Sales Transaction Dataset represents customer purchasing behavior during the Diwali festival season in India. It consists of 11,251 rows and 15 columns, occupying approximately 1.3 MB of memory. Each row represents an individual customer transaction containing demographic attributes such as age, gender, and occupation, along with transactional details including product category, number of orders, and total purchase amount (Kumar and Reinartz, 2016).

2.2. Field Descriptions

Table 1: Field Description Table

Field Name	Description	Business Relevance	Identified Issue
User_ID	Unique identifier for each customer.	Helps track individual customer records.	Identifier only, limited analytical value.
Cust_name	Name of the customer.	No significant business relevance for analysis.	Contains personal information and should be removed.
Product_ID	Unique identifier for each product.	Supports product tracking and inventory analysis.	Limited value for statistical analysis.
Gender	Gender of the customer.	Helps identify gender based purchasing trends.	No major issue.

Age_Group	Age range category of the customer.	Supports customer segmentation by age group.	Requires transformation into simplified categories.
Age	Exact age of the customer.	Helps analyze age-based spending behavior.	Minor missing values.
Marital_Status	Marital status of the customer.	Assists in analyzing family-oriented purchasing patterns.	Needs interpretation for readability.
State	State of residence.	Supports regional sales and demand analysis.	No major issue.
Zone	Geographic zone classification.	Helps identify geographical purchasing patterns.	No major issue.
Occupation	Occupation of the customer.	Supports occupational customer segmentation.	Possible inconsistencies in category labels.
Product_Category	Category of purchased product.	Identifies high-demand and profitable product segments.	No major issue.
Orders	Number of items ordered.	Reflects customer purchasing frequency.	No major issue.
Amount	Total purchase amount.	Represents sales revenue generated per transaction.	Contains missing values.
Status	Empty or null attribute.	No analytical or business relevance.	Mostly null values.
unnamed1	Empty column.	No analytical value.	Completely empty column.

2.3. Identified Issues and Challenges

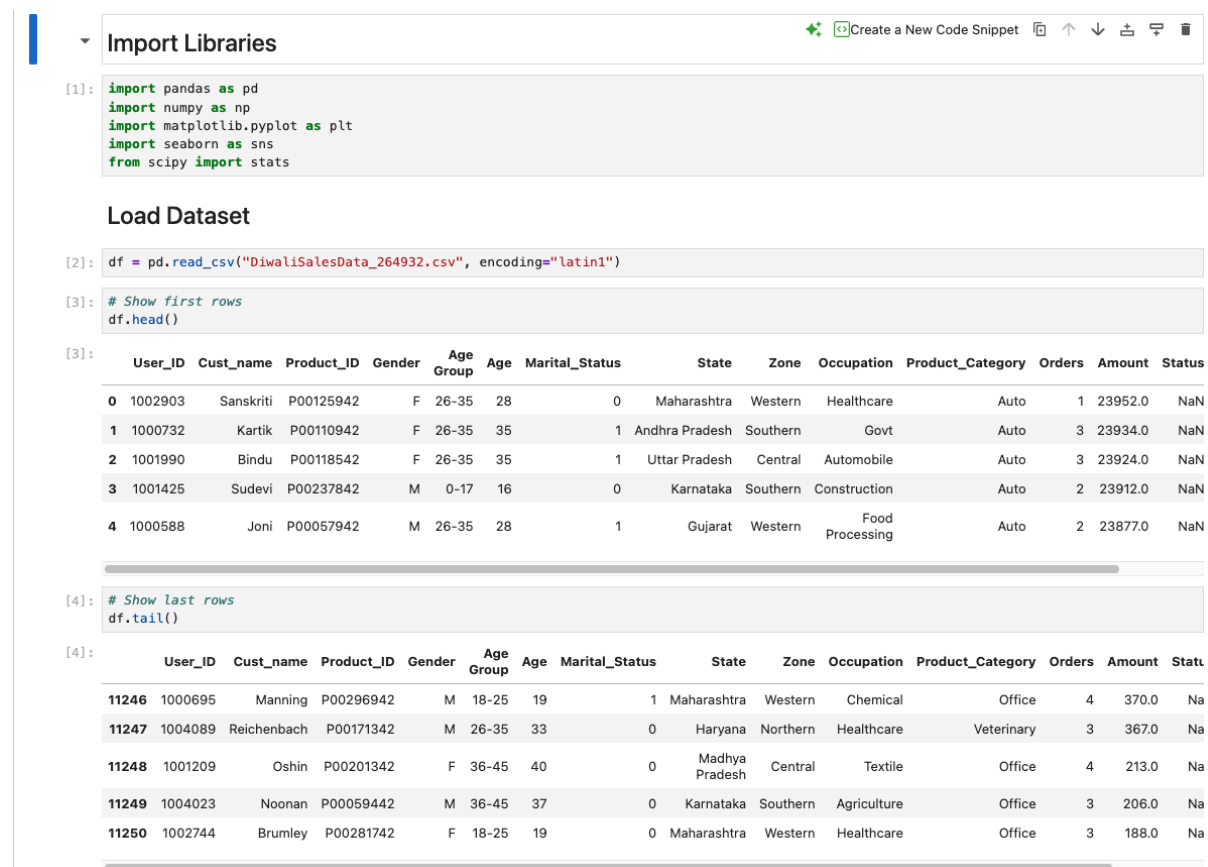
Missing values were identified in the Amount and Age columns, which are critical variables for evaluating sales performance and customer demographics. The Status and unnamed1 columns contained entirely null values with no analytical value, while User_ID and Cust_name function as identifiers irrelevant to statistical analysis. Additionally, the Age_Group variable requires transformation into simplified ordinal categories to support structured comparison across customer segments.

3. Data Preparation

Data preparation involves organizing and structuring raw data to ensure it is suitable for analysis. This stage includes importing the dataset, examining its structure, and identifying key variables required for further preprocessing and analysis.

3.1. Import the Dataset

The dataset was stored in CSV format and imported using the pandas library with latin1 encoding to handle special characters. Additionally, environment libraries including numpy for calculations, matplotlib and seaborn for visualizations, and scipy.stats for statistical analysis were also imported.



Import Libraries

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

Load Dataset

```
[2]: df = pd.read_csv("DiwaliSalesData_264932.csv", encoding="latin1")
[3]: # Show first rows
df.head()
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Category	Orders	Amount	Status
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	Auto	1	23952.0	NaN
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	Auto	3	23934.0	NaN
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	Auto	3	23924.0	NaN
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	Auto	2	23912.0	NaN
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	Auto	2	23877.0	NaN

```
[4]: # Show last rows
df.tail()
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Category	Orders	Amount	Status
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	Office	4	370.0	Na
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	Veterinary	3	367.0	Na
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	Office	4	213.0	Na
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	Office	3	206.0	Na
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	Office	3	188.0	Na

Figure 1: Importing Dataset and Libraries

The head() and tail() outputs confirmed that the dataset loaded successfully with all 15 columns accessible. Initial inspection showed a balanced mix of demographic and transactional variables, providing a solid foundation for both customer segmentation and sales analysis.

3.2. Dataset Insight

The structural composition of the dataset was examined using the `df.info()` method and `df.dtypes` attribute to identify variable data types, non-null counts, and memory usage. Also, `df.isnull().sum()` is used to find out the missing values.

```
# Show dataset information
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   User_ID               11251 non-null  int64
1   Cust_name             11251 non-null  object
2   Product_ID            11251 non-null  object
3   Gender                11251 non-null  object
4   Age Group             11251 non-null  object
5   Age                   11251 non-null  int64
6   Marital_Status        11251 non-null  int64
7   State                 11251 non-null  object
8   Zone                  11251 non-null  object
9   Occupation             11251 non-null  object
10  Product_Category      11251 non-null  object
11  Orders                 11251 non-null  int64
12  Amount                11239 non-null  float64
13  Status                 0 non-null      float64
14  unnamed1               0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB

# Show Columns and Datatypes
df.dtypes

User_ID                int64
Cust_name              object
Product_ID             object
Gender                 object
Age Group              object
Age                    int64
Marital_Status         int64
State                  object
Zone                   object
Occupation             object
Product_Category       object
Orders                 int64
Amount                 float64
Status                 float64
unnamed1               float64
dtype: object
```

Figure 2: Dataset Insight 1

```
# Show missing values
df.isnull().sum()

User_ID                0
Cust_name              0
Product_ID             0
Gender                 0
Age Group              0
Age                    0
Marital_Status         0
State                  0
Zone                   0
Occupation             0
Product_Category       0
Orders                 0
Amount                 12
Status                 11251
unnamed1               11251
dtype: int64
```

Figure 3: Dataset Insight 2

The inspection confirmed that the dataset contains 11,251 records across 15 columns, consisting of integer, float, and object data types, with a total memory footprint of approximately 1.3 MB. The output also revealed missing values within specific features, highlighting structural issues that require cleaning and preprocessing prior to formal statistical analysis.

Table 2: Business Variables Tables

Column	Business Relevance
Amount	Measures sales revenue generated per transaction.
Orders	Indicates customer purchasing frequency.
Product_Category	Identifies high-demand product segments.
State	Supports regional sales analysis.
Occupation	Helps analyze customer spending behavior.
Gender	Supports demographic-based purchasing analysis.
Age_Group	Helps identify spending trends across age segments.

The identified variables provide strong analytical value for evaluating customer purchasing behavior, regional sales performance, and demographic-based spending patterns.

4. Data Cleaning

Data cleaning was performed to improve dataset quality, eliminate noise, and prepare the data for accurate analysis. This process involved segmenting numerical age into categories, building an ordinal variable matching these age groups, dropping redundant columns, handling critical missing values, and validating unique categorical entries for consistency.

4.1. Convert Age Group to Ordinal Variable

The Age_Group column was initially stored as an unordered categorical text variable. To support structured comparison and ordered analysis, the age ranges were consolidated into three ordinal categories: Teen, Adult, and Senior using `pd.Categorical()` with `ordered=True`. This conversion established a logical sequence (Teen < Adult < Senior) suitable for sorting, visualization, and statistical analysis while preventing incorrect alphabetical ordering of categories (Han, Kamber and Pei, 2012).

Age Group into Ordinal Category

```
# Show unique Age Group values before conversion
df["Age_Group"].unique()

array(['26-35', '0-17', '18-25', '55+', '46-50', '51-55', '36-45'],
      dtype=object)

# Map age ranges into three simplified ordinal categories
df["Age_Group_Category"] = df["Age_Group"].replace({
    "0-17": "Teen",
    "18-25": "Adult",
    "26-35": "Adult",
    "36-45": "Adult",
    "46-50": "Senior",
    "51-55": "Senior",
    "55+": "Senior"
})

# Apply pd.Categorical with ordered=True to establish logical sequence Teen < Adult < Senior
df["Age_Group_Category"] = pd.Categorical(
    df["Age_Group_Category"],
    categories=["Teen", "Adult", "Senior"],
    ordered=True
)

# Confirm ordered categories after conversion
print("Age Group Values After Conversion:")
print(df["Age_Group_Category"].unique())

Age Group Values After Conversion:
['Adult', 'Teen', 'Senior']
Categories (3, object): ['Teen' < 'Adult' < 'Senior']

# Map categories to ordinal numbers for numerical analysis
df["Age_Group_Ordinal"] = df["Age_Group_Category"].map({
    "Teen": 1,
    "Adult": 2,
    "Senior": 3
})

# Display age group conversion result
df[["Age_Group", "Age_Group_Category", "Age_Group_Ordinal"]].head()
```

	Age_Group	Age_Group_Category	Age_Group_Ordinal
0	26-35	Adult	2
1	26-35	Adult	2
2	26-35	Adult	2
3	0-17	Teen	1
4	26-35	Adult	2

Figure 4: Age Group into Ordinal Category

Following conversion, the column contained three ordered categories confirmed as Teen, Adult, and Senior respectively. This ordinal structure ensures that age based comparisons follow a meaningful progression during visualization and statistical analysis, rather than being ordered alphabetically.

4.2. Creation of Purchase Value Category

The continuous Amount variable was transformed into grouped spending categories using the `pd.cut()` function. Transaction values between 0-5000 were categorized as Low, 5000-10000 as Medium, and values above 10000 as High based on basic business segmentation logic. This transformation simplified customer spending analysis by converting continuous monetary values into interpretable business segments suitable for comparison and visualization (Provost and Fawcett, 2013).

Purchase Value Category

```
# Categorize Amount into Low, Medium, High based on spending thresholds
# 0-5000 = Low, 5000-10000 = Medium, above 10000 = High
df["Purchase_Value_Category"] = pd.cut(
    df["Amount"],
    bins=[0, 5000, 10000, df["Amount"].max()],
    labels=["Low", "Medium", "High"],
    include_lowest=True
)
```

```
# Display Amount and corresponding Purchase Value Category
df[["Amount", "Purchase_Value_Category"]].head(10)
```

	Amount	Purchase_Value_Category
0	23952.00	High
1	23934.00	High
2	23924.00	High
3	23912.00	High
4	23877.00	High
5	23877.00	High
6	23841.00	High
7	NaN	NaN
8	23809.00	High
9	23799.99	High

```
# Show distribution of Purchase Value Categories
print("Purchase Value Category Distribution:")
print(df["Purchase_Value_Category"].value_counts())
```

```
Purchase Value Category Distribution:
Purchase_Value_Category
Medium    4893
High     4154
Low       2192
Name: count, dtype: int64
```

Figure 5: Purchase Value Category

The resulting distribution confirmed that medium spending transactions were the most frequent with 4,893 records, followed by High with 4,154 and Low with 2,192, indicating that most customers fall within moderate to high spending ranges.

4.3. Removal of Irrelevant Columns

Columns including User_ID, Cust_name, Status, and unnamed1 were removed to improve analytical relevance and reduce dataset complexity. User_ID and Cust_name function as identifiers and do not contribute to statistical analysis, while Status and unnamed1 contain mostly null values with no meaningful business information.

Removing Irrelevant Columns

```
# Show all columns before dropping
print("Columns Before Dropping:")
df.columns

Columns Before Dropping:
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount', 'Status', 'unnamed1', 'Age_Group_Category',
       'Age_Group_Ordinal', 'Purchase_Value_Category'],
      dtype='object')

# Drop irrelevant columns
# User_ID and Cust_name are identifiers not useful for analysis
# Status and unnamed1 contain entirely null values
df = df.drop(columns=["Cust_name", "User_ID", "unnamed1", "Status"])

# Confirm columns remaining after dropping
print("Columns After Dropping:")
df.columns

Columns After Dropping:
Index(['Product_ID', 'Gender', 'Age Group', 'Age', 'Marital_Status', 'State',
       'Zone', 'Occupation', 'Product_Category', 'Orders', 'Amount',
       'Age_Group_Category', 'Age_Group_Ordinal', 'Purchase_Value_Category'],
      dtype='object')
```

Figure 6: Removing Irrelevant Column

Following removal, the dataset retained 11 columns suitable for analysis. The removal of those four columns reduced the dataset from 15 to 11 columns, eliminating noise and ensuring that only analytically relevant variables were retained for subsequent analysis.

4.4. Handling Missing Value

Missing values were identified primarily within the Amount and Age column. Common handling techniques include mean imputation, median imputation, forward fill, backward fill, and row removal. In this analysis, rows containing missing values in critical variables such as Amount and Age were removed using `dropna()` because imputing values into the primary sales variable could introduce analytical bias and reduce result reliability (Field, 2013).

Managing Missing Values

```
# Show missing values in critical columns before removal
print("Missing Values Before Removal:")
df[["Amount", "Age"]].isnull().sum()

Missing Values Before Removal:
Amount    12
Age        0
dtype: int64

# Remove rows with missing values in critical columns Amount and Age
# Row removal chosen over imputation to avoid introducing bias into primary sales variable
df = df.dropna(subset=["Amount", "Age"])

# Confirm missing values removed
print("Missing Values After Removal:")
df[["Amount", "Age"]].isnull().sum()

Missing Values After Removal:
Amount     0
Age         0
dtype: int64

# Final check - confirm no missing values remain across any column
print("Final Missing Value Check:")
df.isnull().sum()

Final Missing Value Check:
Product_ID      0
Gender          0
Age Group       0
Age             0
Marital_Status  0
State           0
Zone            0
Occupation      0
Product_Category 0
Orders          0
Amount          0
Age_Group_Category 0
Age_Group_Ordinal 0
Purchase_Value_Category 0
dtype: int64

# Confirm final dataset size after cleaning
print("Final Dataset Shape:", df.shape)

Final Dataset Shape: (11239, 14)
```

Figure 7: Managing Missing Values

A total of 12 rows were removed, reducing the dataset from 11,251 to 11,239 records, after which no missing values remained across any column. The removal of only 12 rows represents a minimal data loss of approximately 0.1% of the total dataset, ensuring that analytical integrity was maintained without significantly affecting sample size.

4.5. Identification of Unique Values

Unique categorical values were examined to understand dataset structure and identify inconsistencies within category labels. The analysis identified 5 geographical zones (Central, Eastern, Northern, Southern, and Western) and 18 unique Product_Category values. Examining unique distributions helps detect duplicate labels, inconsistent formatting, or unexpected entries that may negatively affect analytical accuracy (Han, Kamber and Pei, 2012).

Unique Values

```
# Display unique values for all key categorical columns
categorical_columns = [
    "Gender",
    "Age_Group_Category",
    "Purchase_Value_Category",
    "Marital_Status",
    "State",
    "Zone",
    "Occupation",
    "Product_Category"
]

for column in categorical_columns:
    print(f"{column}: ({df[column].nunique()} unique values):")
    print(df[column].unique())
    print()
```

Gender (2 unique values):
['F' 'M']

Age_Group_Category (3 unique values):
['Adult', 'Teen', 'Senior']
Categories (3, object): ['Teen' < 'Adult' < 'Senior']

Purchase_Value_Category (3 unique values):
['High', 'Medium', 'Low']
Categories (3, object): ['Low' < 'Medium' < 'High']

Marital_Status (2 unique values):
[0 1]

State (16 unique values):
['Maharashtra' 'Andhra Pradesh' 'Uttar Pradesh' 'Karnataka' 'Gujarat'
'Himachal Pradesh' 'Delhi' 'Jharkhand' 'Kerala' 'Haryana'
'Madhya Pradesh' 'Bihar' 'Rajasthan' 'Uttarakhand' 'Telangana' 'Punjab']

Zone (5 unique values):
['Western' 'Southern' 'Central' 'Northern' 'Eastern']

Occupation (15 unique values):
['Healthcare' 'Govt' 'Automobile' 'Construction' 'Food Processing'
'Lawyer' 'Media' 'Banking' 'Retail' 'IT Sector' 'Aviation' 'Hospitality'
'Agriculture' 'Textile' 'Chemical']

Product_Category (18 unique values):
['Auto' 'Hand & Power Tools' 'Stationery' 'Tupperware' 'Footwear & Shoes'
'Furniture' 'Food' 'Games & Toys' 'Sports Products' 'Books'
'Electronics & Gadgets' 'Decor' 'Clothing & Apparel' 'Beauty'
'Household items' 'Pet Care' 'Veterinary' 'Office']

Figure 8: Unique Values

No unexpected labels or inconsistent formatting were detected across any categorical column, confirming that the dataset was clean and consistent for analysis. The Gender column confirmed two categories (F and M), Occupation contained 15 unique values, and Marital_Status contained two values (0 and 1) representing Single and Married customers respectively, all consistent with expected dataset structure.

5. Data Analysis

Data analysis involves examining and interpreting data to identify meaningful patterns, trends, and relationships. In this study, summary statistics and correlation analysis were applied to evaluate customer purchasing behavior and relationships between numerical variables.

5.1. Summary Statistics

Summary statistics were calculated to understand the distribution and variability of numerical variables within the dataset (Field, 2013).

- Mean: Represents the average value of the data.
- Median: Represents the middle value of the dataset.
- Mode: Represents the most frequently occurring value.
- Standard Deviation: Measures the spread or dispersion of data values.
- Skewness: Identifies the symmetry of the distribution.
- Kurtosis: Measures the shape and peaked of the distribution.

Summary Statistics

```
# Compute mean, standard deviation, skewness, and kurtosis for numerical variables
summary_statistics = df[["Age", "Orders", "Amount"]].agg(["mean", "median", "std", "skew", "kurt"])
summary_statistics
```

	Age	Orders	Amount
mean	35.410357	2.489634	9453.610858
median	33.000000	2.000000	8109.000000
std	12.753866	1.114967	5222.355869
skew	1.185478	0.019284	0.558026
kurt	2.472002	-1.352541	-0.540209

```
# Compute mode for numerical variables
print("Mode:")
df[["Age", "Orders", "Amount"]].mode().iloc[0]
```

```
Mode:
Age      30.0
Orders    2.0
Amount   7907.0
Name: 0, dtype: float64
```

Figure 9: Summary Statistics 1

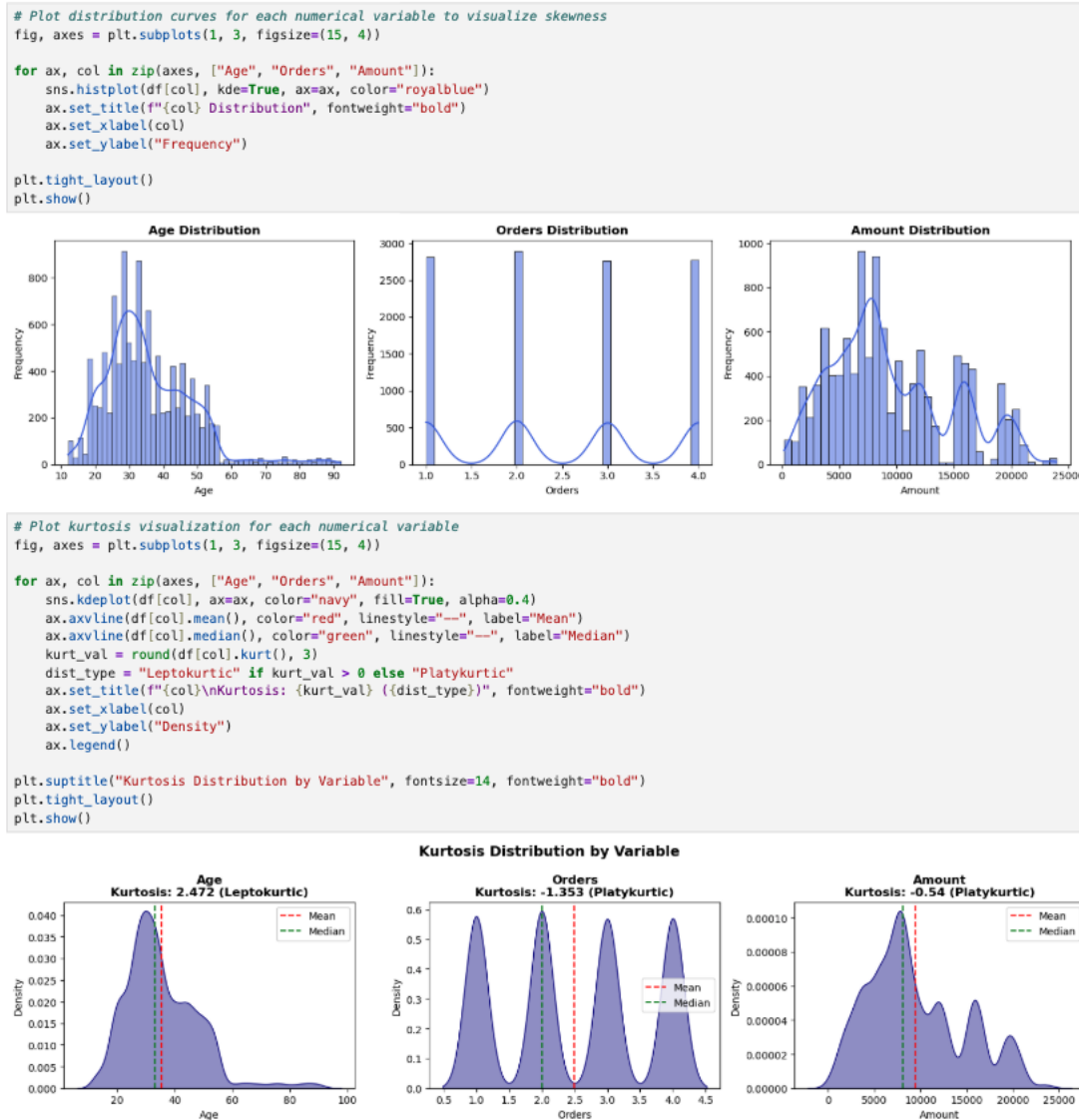


Figure 10: Summary Statistic 2

The Amount variable recorded a mean of Rs.9,453.61 and standard deviation of Rs.5,222.36, indicating high variability in customer spending behavior. Positive skewness of 0.558 suggested that most transactions were concentrated below the average with fewer high-value purchases. The Age variable recorded a mean of 35.41 years with slight positive skewness of 1.185 and positive kurtosis of 2.472, indicating a leptokurtic distribution with the presence of some extreme values. Orders showed a mean of 2.49 with near-zero skewness and kurtosis of -1.353, indicating a relatively symmetrical and platykurtic distribution. In contrast, amount recorded kurtosis of -0.540, suggesting a positively skewed and platykurtic distribution where most purchases remain below the average with a smaller number of high-value transactions.

5.2. Correlation Analysis

Correlation analysis was performed to measure the strength and direction of relationships between numerical variables. A correlation matrix summarizes pairwise correlation coefficients, while a heatmap visualizes them using color intensity (Field, 2013). Pearson correlation measures linear relationships, whereas Spearman correlation measures monotonic relationships suitable for non-linear data (Schober, Boer and Schwarte, 2018). Correlation coefficients range from -1 to +1, where +1 indicates a perfect positive relationship, -1 indicates a perfect negative relationship, and 0 indicates no relationship.

Correlation Analysis

```
# Compute Pearson correlation matrix
pearson_matrix = df[["Age", "Orders", "Amount"]].corr(method="pearson")
print("Pearson Correlation Matrix:")
pearson_matrix
```

Pearson Correlation Matrix:

	Age	Orders	Amount
Age	1.000000	0.008090	0.030941
Orders	0.008090	1.000000	-0.013183
Amount	0.030941	-0.013183	1.000000

```
# Compute Spearman correlation matrix for monotonic relationships
spearman_matrix = df[["Age", "Orders", "Amount"]].corr(method="spearman")
print("Spearman Correlation Matrix:")
spearman_matrix
```

Spearman Correlation Matrix:

	Age	Orders	Amount
Age	1.000000	0.007047	0.038446
Orders	0.007047	1.000000	-0.013389
Amount	0.038446	-0.013389	1.000000

```
# Print specific pairwise correlation values
print("Age vs Amount (Pearson r):", round(pearson_matrix.loc["Age", "Amount"], 4))
print("Orders vs Amount (Pearson r):", round(pearson_matrix.loc["Orders", "Amount"], 4))
```

Age vs Amount (Pearson r): 0.0309
Orders vs Amount (Pearson r): -0.0132

Figure 11: Correlation Analysis 1

```
# Print specific pairwise correlation values
print("Age vs Amount (Pearson r):", round(pearson_matrix.loc["Age", "Amount"], 4))
print("Orders vs Amount (Pearson r):", round(pearson_matrix.loc["Orders", "Amount"], 4))

Age vs Amount (Pearson r): 0.0309
Orders vs Amount (Pearson r): -0.0132
```

```
# Visualize Pearson and Spearman correlation matrices as subplots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
# Pearson Heatmap
sns.heatmap(pearson_matrix, annot=True, cmap="coolwarm", fmt=".2f",
            linewidths=0.5, ax=axes[0])
axes[0].set_title("Pearson Correlation Matrix", fontsize=13, fontweight="bold")
# Spearman Heatmap
sns.heatmap(spearman_matrix, annot=True, cmap="viridis", fmt=".2f",
            linewidths=0.5, ax=axes[1])
axes[1].set_title("Spearman Correlation Matrix", fontsize=13, fontweight="bold")
plt.suptitle("Correlation Analysis", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.show()
```

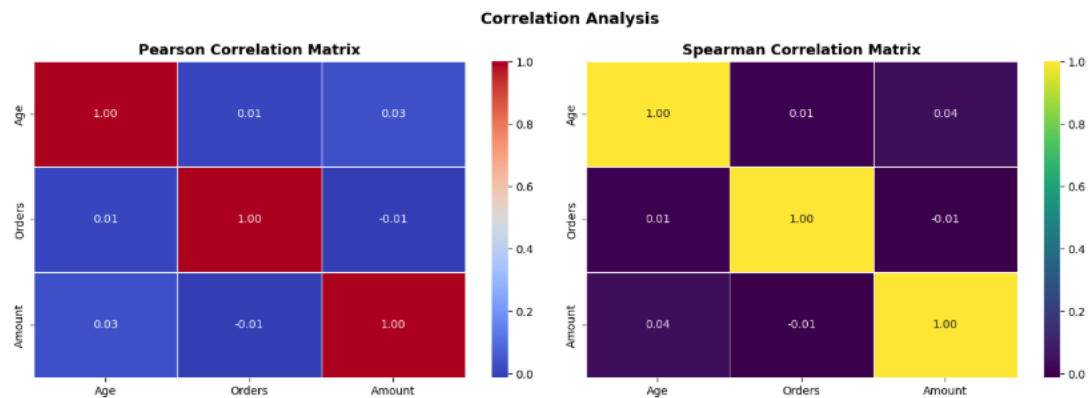


Figure 12: Correlation Analysis 2

The results revealed weak relationships among all variables. Age and Amount showed a very weak positive correlation ($r = 0.03$), while Orders and Amount showed the weakest correlation ($r = -0.01$). Spearman correlation produced similar results, confirming that the weak relationships are not dependent on linearity assumptions. Overall, this indicates that purchasing behavior is influenced by multiple independent factors rather than a single dominant variable.

6. Data Exploration

Data exploration was performed using visual analysis techniques including bar charts, grouped bar charts, and crosstab visualizations to identify trends, patterns, and relationships within the dataset. These techniques were applied to examine regional sales performance, demographic purchasing behavior, and category-based spending patterns (Kumar and Reinartz, 2016).

6.1. Visualization of Insights

6.1.1. Top 5 states by Total Amounts

The dataset was grouped by State to identify the highest revenue-generating regions. Uttar Pradesh recorded the highest total purchase amount at approximately Rs.19.37M, followed by Maharashtra at Rs.14.43M and Karnataka at Rs.13.52M. The results indicate that customer purchasing activity is concentrated within a few major states, reflecting regional variation in sales performance.

Top 5 States by Total Amount

```
# Group by State and sum total Amount, select top 5
top_states = df.groupby("State")["Amount"].sum().sort_values(ascending=False).head(5)
print("Top 5 States by Total Amount:")
top_states
```

Top 5 States by Total Amount:

```
State
Uttar Pradesh    19374968.00
Maharashtra     14427543.00
Karnataka        13523540.00
Delhi            11603819.45
Madhya Pradesh   8101142.00
Name: Amount, dtype: float64
```

```
# Bar chart for Top 5 States by Total Amount
plt.figure(figsize=(10, 6))
bars = plt.bar(top_states.index, top_states.values, color="indigo")
plt.title("Top 5 States by Total Amount", fontsize=14, fontweight="bold")
plt.xlabel("State")
plt.ylabel("Total Amount (Rs.)")
plt.xticks(rotation=45)
plt.gca().yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{x:,.0f}'))
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```

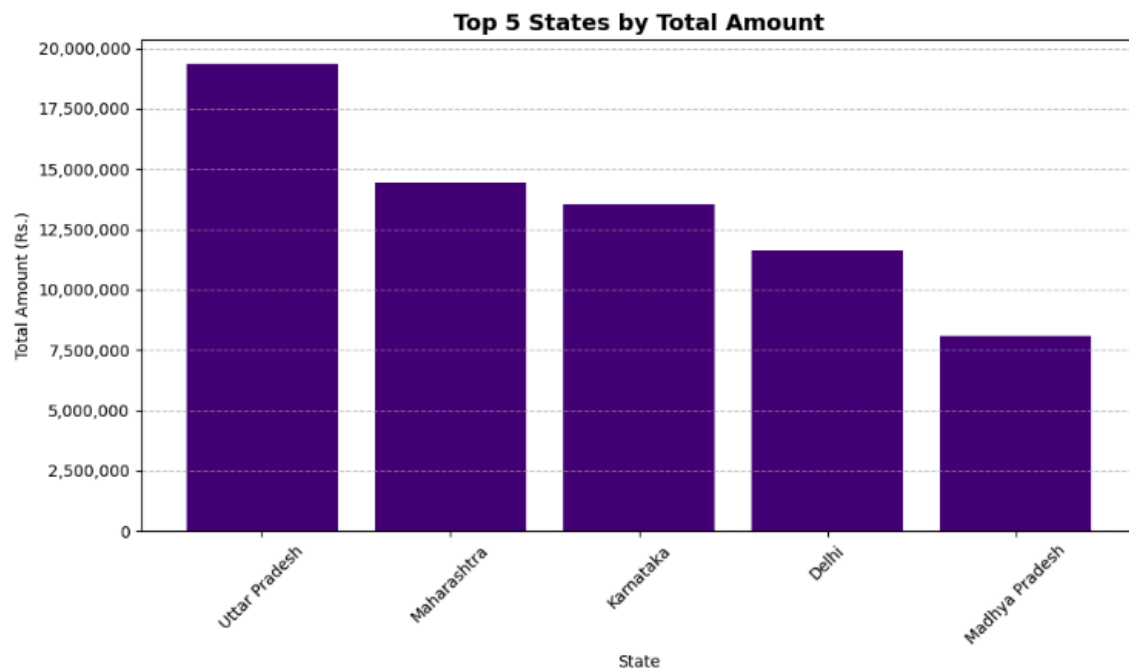


Figure 13: Top 5 States by Total Amount

6.1.2. Distribution of Gender across Product Category

Gender distribution was analyzed across product categories to identify purchasing preferences between male and female customers. Clothing & Apparel recorded the highest female participation with 1,854 female customers compared to 801 male customers. In contrast, male customers were more concentrated in categories including Sports Products and Auto. Categories such as Food and Electronics & Gadgets showed relatively balanced gender participation, while Clothing & Apparel, Beauty, and Footwear & Shoes were heavily female dominated. These differences suggest that purchasing preferences vary significantly across demographic groups.

Distribution of Gender across Product Category

```
# Compute gender distribution across product categories using crosstab
gender_product = pd.crosstab(df["Product_Category"], df["Gender"])
gender_product
```

	Gender	
Product_Category	F	M
Auto	64	33
Beauty	325	97
Books	39	64
Clothing & Apparel	1854	801
Decor	79	17
Electronics & Gadgets	1476	611
Food	1771	719
Footwear & Shoes	782	277
Furniture	219	133
Games & Toys	273	113
Hand & Power Tools	9	17
Household Items	284	236
Office	75	38
Pet Care	144	68
Sports Products	264	92
Stationery	59	53
Tupperware	52	20
Veterinary	63	18

Figure 14: Distribution of Gender across Product Category 1

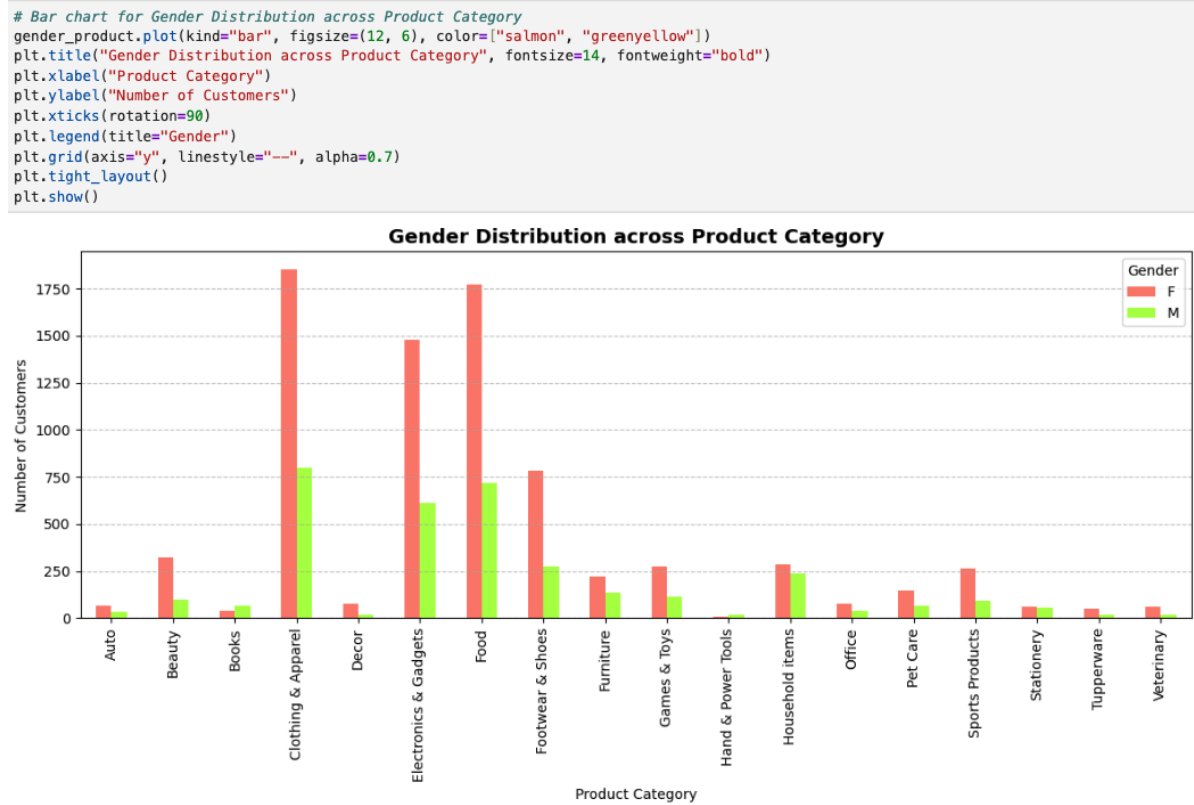


Figure 15: Distribution of Gender across Product Category 2

6.1.3. Age Group vs Average Amount

Average purchase amount was calculated for each age group to evaluate spending behavior across customer segments. Senior customers recorded the highest average spending at approximately Rs.9,620, followed by Adults at Rs.9,421, while Teen customers recorded the lowest average purchase value at Rs.9,120. This suggests that purchasing power generally increases with age.

Age Group vs Average Amount

```
# Calculate average purchase amount per age group
age_amount = df.groupby("Age_Group_Category", observed=True)["Amount"].mean()
print("Average Amount by Age Group:")
age_amount
```

Average Amount by Age Group:

```
Age_Group_Category
Teen      9120.449324
Adult     9421.943172
Senior    9620.673214
Name: Amount, dtype: float64
```

```
# Bar chart for Age Group vs Average Amount
plt.figure(figsize=(8, 5))
bars = plt.bar(age_amount.index, age_amount.values, color="purple")
plt.title("Age Group vs Average Amount", fontsize=14, fontweight="bold")
plt.xlabel("Age Group")
plt.ylabel("Average Amount (Rs.)")
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```

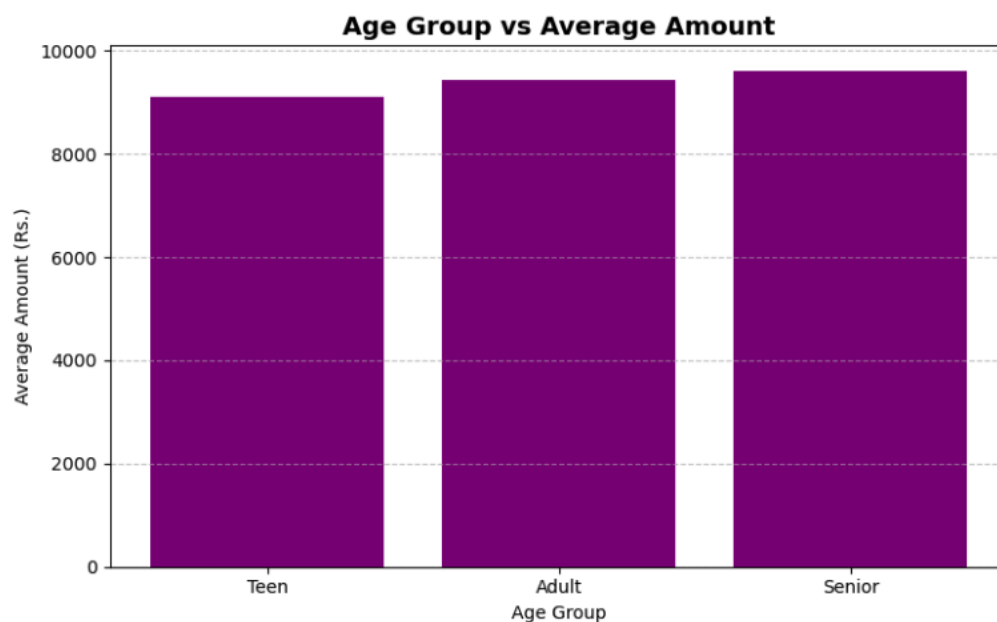


Figure 16: Age Group vs Average Amount

6.1.4. Marital Status Impact on Orders

The relationship between marital status and purchasing frequency was examined using average order counts. Single customers recorded a slightly higher average of 2.493 orders compared to married customers at 2.485, indicating marginal difference in purchasing frequency between the two groups. For improved readability, marital status categories were represented as Single and Married instead of numerical values (0 and 1) within the visualization.

Marital Status Impact on Orders

```
# Calculate average orders per marital status group
marital_orders = df.groupby("Marital_Status")["Orders"].mean()
# Map 0 and 1 to readable labels
marital_orders.index = marital_orders.index.map({0: "Single", 1: "Married"})
print("Average Orders by Marital Status:")
marital_orders
```

Average Orders by Marital Status:

```
Marital_Status
Single      2.492943
Married     2.485067
Name: Orders, dtype: float64
```

```
# Bar chart for Marital Status Impact on Orders
plt.figure(figsize=(8, 5))
bars = plt.bar(marital_orders.index, marital_orders.values, color="tomato")
plt.title("Marital Status Impact on Orders", fontsize=14, fontweight="bold")
plt.xlabel("Marital Status")
plt.ylabel("Average Orders")
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```

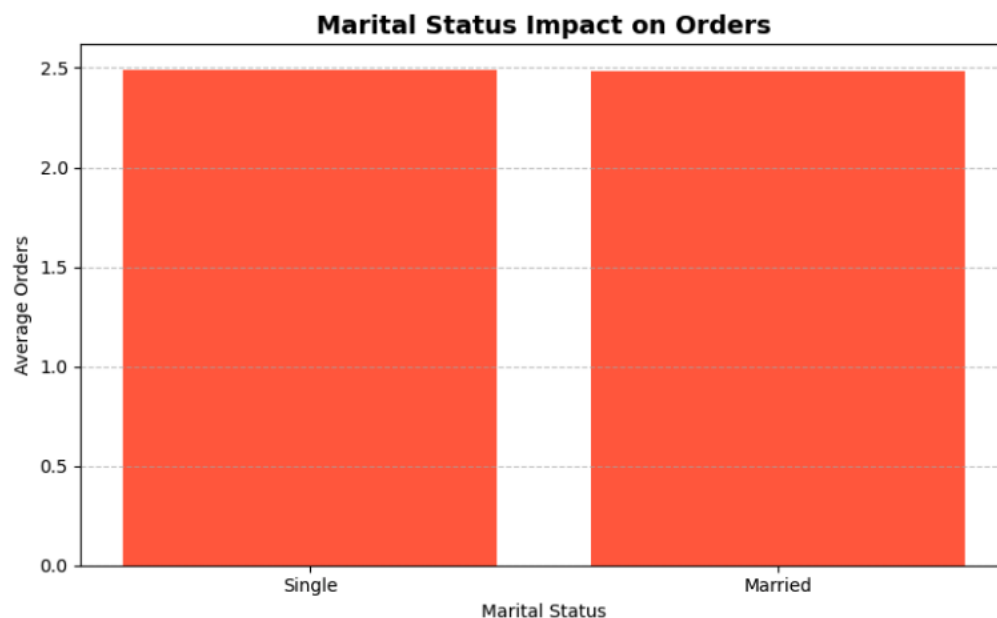


Figure 17: Marital Status Impact on Order

6.2. Categorized Analysis

Categorized analysis was performed using grouped visualizations to compare average purchase amounts across product categories and geographical zones. The results showed that categories such as Food, Clothing & Apparel, and Electronics & Gadgets consistently generated higher average spending across multiple zones. For instance, the Auto category recorded the highest average purchase amount in the Central zone at approximately Rs.20,705.

Categorized Analysis

```
# Group by Zone and Product_Category to compute average Amount
zone_product = df.groupby(["Zone", "Product_Category"])["Amount"].mean().unstack()
zone_product
```

Product_Category	Auto	Beauty	Books	Clothing & Apparel	Decor	Electronics & Gadgets	Food	Footwear & Shoes	Furniture	Ga
Zone										
Central	20705.100000	4626.664495	10269.892857	6284.684902	7823.558824	7485.329502	13865.816109	14754.340505	15660.439394	11621.0
Eastern	19908.000000	6022.000000	10483.000000	6154.062500	NaN	7456.041096	13311.909639	16606.263158	15119.000000	10482.1
Northern	21394.000000	4163.300000	10838.153846	6047.326437	NaN	7501.695489	13840.436620	14162.100000	15843.686275	11074.2
Southern	19958.919194	4737.177419	10190.702128	6313.712803	NaN	7509.591876	13517.455720	14653.649596	15804.255976	11506.5
Western	20263.857143	4642.350000	10238.153846	6133.031818	7084.214286	7509.326599	13340.329897	14297.415385	14425.626866	10761.3

Figure 18: Categorized Analysis 1



Figure 19: Categorized Analysis 2

The rankings of product categories varied across geographical regions, indicating that customer preferences differ between zones. These variations suggest that regional purchasing behavior influences product demand and may support targeted marketing and inventory planning strategies.

7. Statistical Testing

Statistical testing was performed to determine whether observed patterns within the dataset were statistically significant or occurred by chance. Common techniques include t-tests, ANOVA, and Chi-square tests, where t-tests compare means between two groups, ANOVA compares means across multiple groups, and Chi-square evaluates relationships between categorical variables. These techniques support evidence based decision making and validate analytical findings (Field, 2013).

7.1. ANOVA Test (Occupation Vs Amount)

A one-way ANOVA test was performed to determine whether average purchase amounts differ across occupation groups.

- H_0 : Mean purchase amount is equal across all occupation groups.
- H_A : Mean purchase amount differs across occupation groups.

ANOVA Test (Occupation vs Amount)

```
# Show mean Amount per occupation group for context
print("Mean Amount by Occupation:")
df.groupby("Occupation")["Amount"].mean().sort_values(ascending=False)
```

Mean Amount by Occupation:

Occupation	
Govt	9973.316159
Media	9883.568273
Chemical	9791.933457
Food Processing	9623.333333
Aviation	9620.074809
Retail	9547.245509
Automobile	9501.939823
Banking	9472.832850
Lawyer	9381.666667
IT Sector	9320.959570
Healthcare	9257.519524
Textile	9183.300860
Agriculture	9162.851590
Hospitality	9070.277383
Construction	8689.640097

Name: Amount, dtype: float64

```
# H0: Mean Amount is equal across all Occupation groups
# HA: Mean Amount differs significantly across Occupation groups
# Collect Amount values per occupation group
occupation_groups = []
for occupation, group_data in df.groupby("Occupation"):
    occupation_groups.append(group_data["Amount"])
# Run one-way ANOVA test
f_statistic, anova_p_value = stats.f_oneway(*occupation_groups)
# F-statistic measures variance between groups relative to variance within groups
# Higher F-value indicates greater differences between group means
print("F-statistic:", f_statistic)
print("P-value:", anova_p_value)
```

F-statistic: 2.477069238682129
P-value: 0.001659146529704751

```
# Evaluate result against significance level of 0.05
if anova_p_value < 0.05:
    print("Reject H0")
    print("Conclusion: Mean Amount differs significantly across Occupation groups.")
else:
    print("Fail to Reject H0")
    print("Conclusion: Mean Amount is not significantly different across Occupation groups.")
```

Reject H0
Conclusion: Mean Amount differs significantly across Occupation groups.

Figure 20: ANOVA Test

The test produced an F-statistic of 2.477, which measures variation between occupation groups relative to variation within groups, and a p-value of 0.00165, which is below the significance level of 0.05. The null hypothesis was therefore rejected, indicating that occupation significantly influences customer spending behavior. Among occupation groups, Government employees recorded the highest average purchase amount at Rs.9,973, while Construction workers recorded the lowest at Rs.8,689, indicating notable spending variation across occupational segments.

7.2. Chi-Square Test (Product Category Vs Zone)

A Chi-square test of independence was conducted to examine the relationship between product category and geographical zone.

- H_0 : Product category and zone are independent.
- H_A : Product category and zone are associated.

Chi-Square Test

```
# H0: Product_Category and Zone are independent
# HA: Product_Category and Zone are associated
# Build contingency table of observed frequencies
contingency_table = pd.crosstab(df["Product_Category"], df["Zone"])
contingency_table
```

	Zone	Central	Eastern	Northern	Southern	Western
Product_Category						
Auto		20	4	4	62	7
Beauty		307	3	10	62	40
Books		28	2	13	47	13
Clothing & Apparel		914	288	435	578	440
Decor		68	0	0	0	28
Electronics & Gadgets		522	219	532	517	297
Food		987	166	213	542	582
Footwear & Shoes		574	19	30	371	65
Furniture		132	20	51	82	67
Games & Toys		119	29	60	87	91
Hand & Power Tools		14	0	3	7	2
Household items		176	15	31	172	126
Office		79	5	5	11	13
Pet Care		77	16	37	45	37
Sports Products		102	21	49	83	101
Stationery		85	1	5	5	16
Tupperware		20	3	6	21	22
Veterinary		65	3	7	1	5

```
# Run Chi-square test of independence
chi2_statistic, chi_p_value, degrees_of_freedom, expected_values = stats.chi2_contingency(contingency_table)
# Chi-square statistic measures difference between observed and expected frequencies
# Degrees of freedom = (rows - 1) x (columns - 1)
print("Chi-square statistic:", chi2_statistic)
print("P-value:", chi_p_value)
print("Degrees of Freedom:", degrees_of_freedom)
```

```
Chi-square statistic: 1634.9668637157445
P-value: 1.4536117678958916e-296
Degrees of Freedom: 68
```

Figure 21: Chi-Square Test 1

```
# Display expected values table for comparison with observed frequencies
expected_table = pd.DataFrame(
    expected_values,
    index=contingency_table.index,
    columns=contingency_table.columns
)
print("Expected Values Table:")
expected_table
```

Expected Values Table:

	Zone	Central	Eastern	Northern	Southern	Western
Product_Category						
Auto		37.016905	7.025358	12.868316	23.242370	16.847050
Beauty		161.042619	30.563929	55.983806	101.116291	73.293354
Books		39.306611	7.459916	13.664294	24.680043	17.889136
Clothing & Apparel		1013.194679	192.292019	352.220393	636.170033	461.122876
Decor		36.635288	6.952932	12.735653	23.002758	16.673370
Electronics & Gadgets		796.435893	151.153839	276.867782	500.070380	362.472106
Food		950.227778	180.341667	330.330990	596.634042	432.465522
Footwear & Shoes		404.133019	76.699528	140.490168	253.749177	183.928107
Furniture		134.329389	25.494083	46.697393	84.343447	61.135688
Games & Toys		147.304387	27.956580	51.207937	92.490257	67.040840
Hand & Power Tools		9.922057	1.883086	3.449239	6.229914	4.515704
Household items		198.441142	37.661714	68.984785	124.598274	90.314085
Office		43.122787	8.184180	14.990924	27.076163	19.625945
Pet Care		80.902927	15.354391	28.124566	50.797758	36.820358
Sports Products		135.855859	25.783789	47.228045	85.301895	61.830412
Stationery		42.741169	8.111754	14.858261	26.836551	19.452264
Tupperware		27.476466	5.214699	9.551739	17.252069	12.505027
Veterinary		30.911024	5.866536	10.745707	19.408577	14.068156

```
# Evaluate result against significance level of 0.05
if chi_p_value < 0.05:
    print("Reject H0")
    print("Conclusion: Product_Category and Zone are associated.")
else:
    print("Fail to Reject H0")
    print("Conclusion: Product_Category and Zone are independent.")
```

Reject H0

Conclusion: Product_Category and Zone are associated.

Figure 22: Chi-Square Test 2

The Chi-square statistic was 1634.97 with 68 degrees of freedom and a p-value below 0.05, resulting in rejection of the null hypothesis. Degrees of freedom represent the number of independent category combinations within the contingency table. This confirms that product category and geographical zone are significantly associated, indicating that customer purchasing preferences vary across different regions. Clothing & Apparel showed the largest difference between observed and expected frequencies, suggesting it is the most regionally concentrated product category. This further confirms that geographical zone significantly influences customer product preferences.

8. Conclusion

This project analyzed customer purchasing behavior using the Diwali Sales dataset through data cleaning, statistical analysis, and visualization techniques. Uttar Pradesh recorded the highest total sales exceeding ₹19 million, while Food, Clothing & Apparel, and Electronics & Gadgets consistently showed strong activity across multiple regions. Summary statistics revealed high variability and positive skewness in Amount, while correlation analysis confirmed weak relationships among numerical variables. The ANOVA test confirmed occupation significantly influences spending ($p=0.00165$), while the Chi-square test confirmed a significant association between product category and zone ($\chi^2=1634.97$, $p<0.05$). These findings support targeted marketing, regional sales strategies, and customer segmentation, though the single-period dataset and absence of income data limit deeper behavioral analysis.

9. References

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